AIRLOG Generator

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AIRLOG Generator

Product Overview

AIRLOG Generator was written to allow users to generate air log files from their WideOrbit Automation for Radio Central Server. AIRLOG Generator allows users to generate air log files on demand or based on a user-defined schedule. This application was designed to be run either as a service or a standalone application. AIRLOG Generator provides a browser-based user interface with a built-in web host. As a service, this application will run scheduled tasks based on the individual station configuration json files. It is possible to manually create these files if UI access is not available or use the UI to create the schedule.

Application Disclaimer: This tool was written with the help of Artificial Intelligence.

Windows Installation

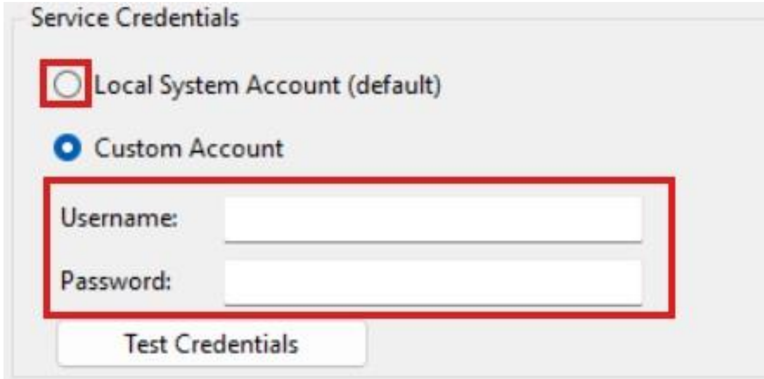
Getting Started

AIRLOG Generator can be installed as a service or run as a standalone application. The provided software package contains AIRLOGGeneratorInstaller.exe that is used to install a service in Windows. The service will be starting with AIRLOGGenerator.exe, the same file used to operate the application in a standalone manner. The application is designed to be self-contained and will operate from any location where the software package is placed.

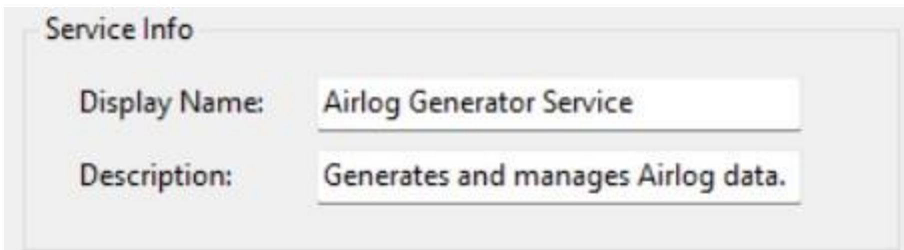
Installing as A Service

In the provided software package, locate the AirlogGeneratorInstaller.exe. Right click on this file and "Run as Administrator".

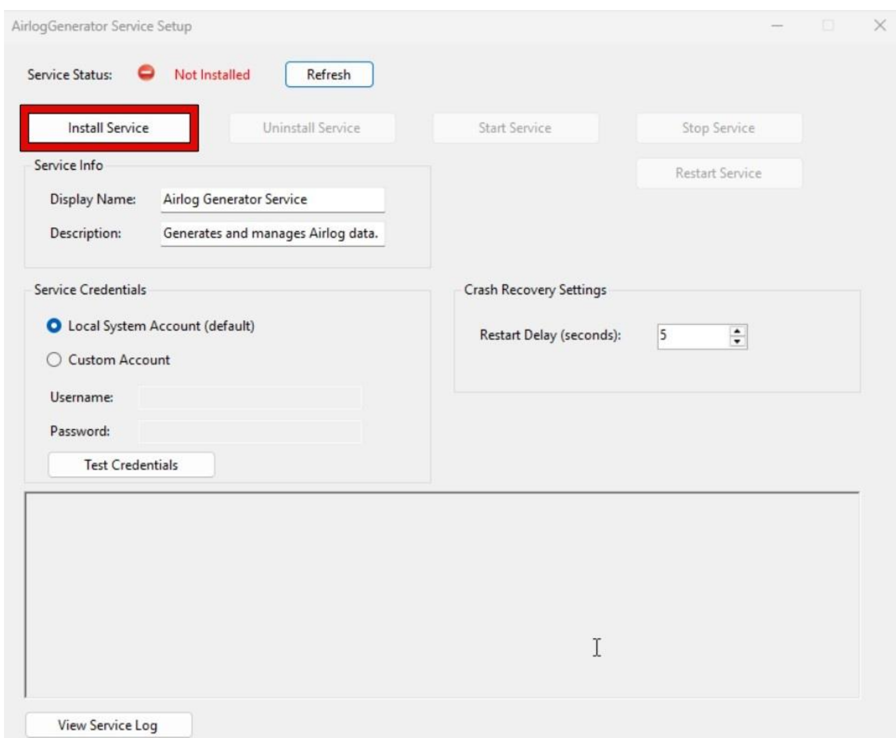
1) By default, the installer is set to use the Local Account for the service
    a. This is the account that will be writing the air logs to disk after generation
    b. It is possible to set a service account if needed



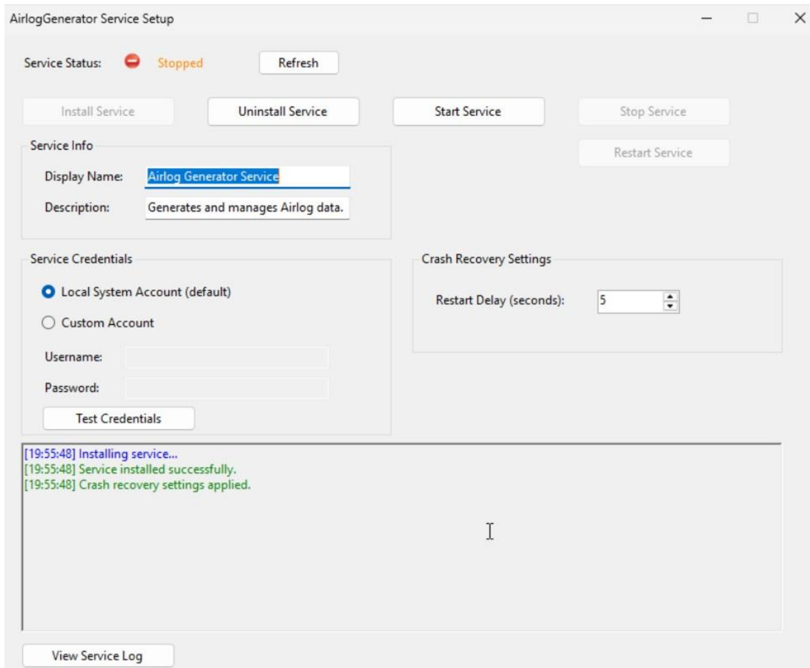
2) By default, the service will be installed as "Airlog Generator Service". It is possible to set this service name to something different if desired.



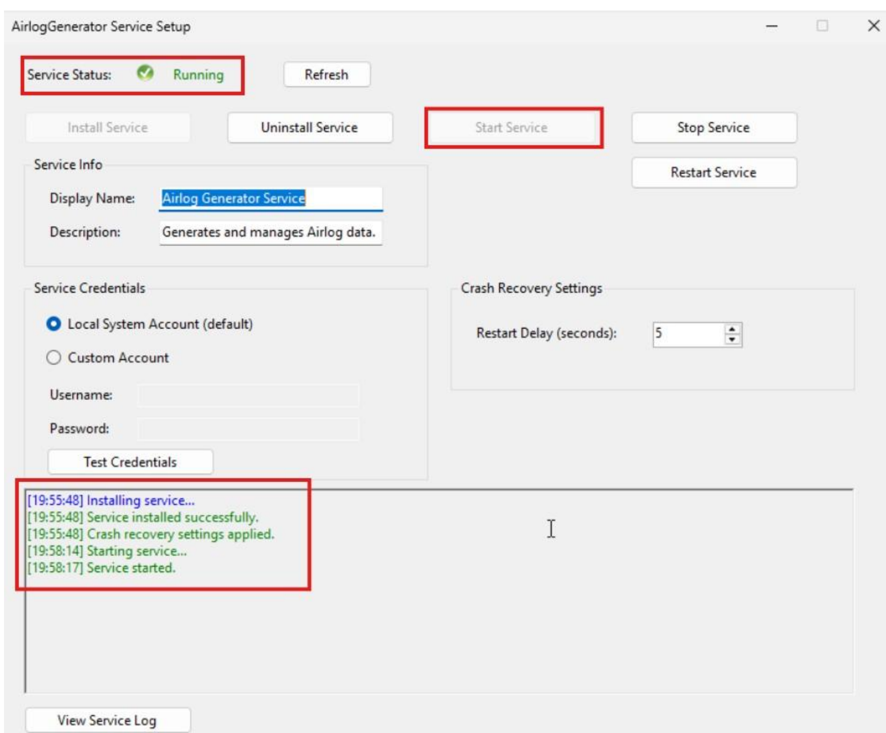
3) Click on the Install Button



4) When the installation is complete. The integrated log window will display a message of success.

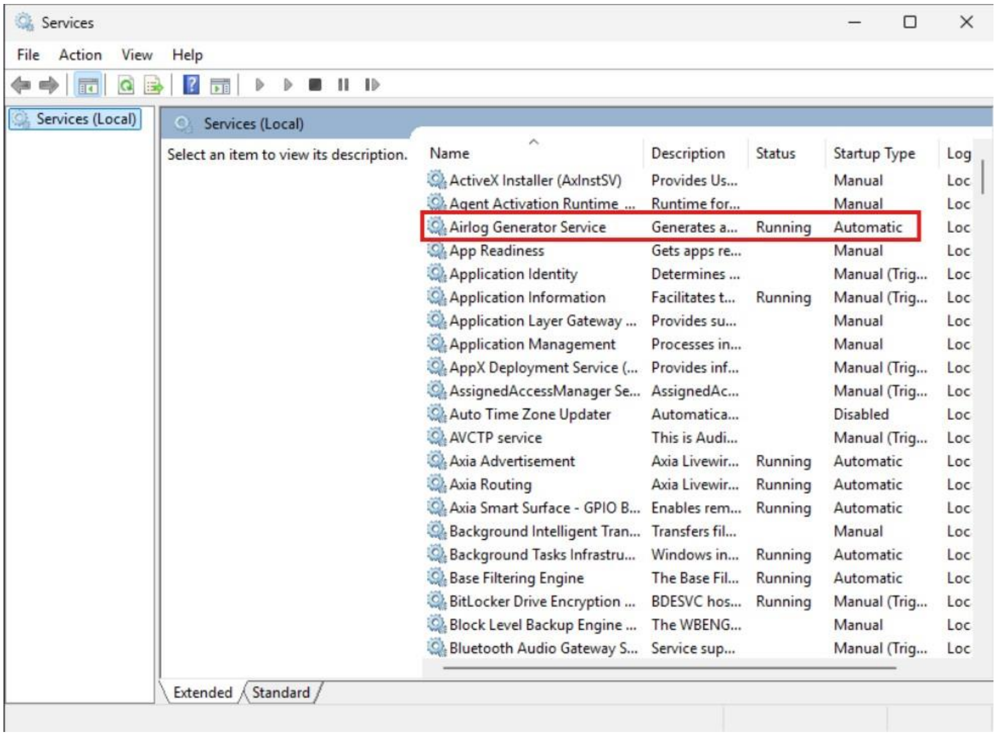


5) After the service is installed, start the service.



6) The service can be verified in the Services window.

AIRLOG Generator

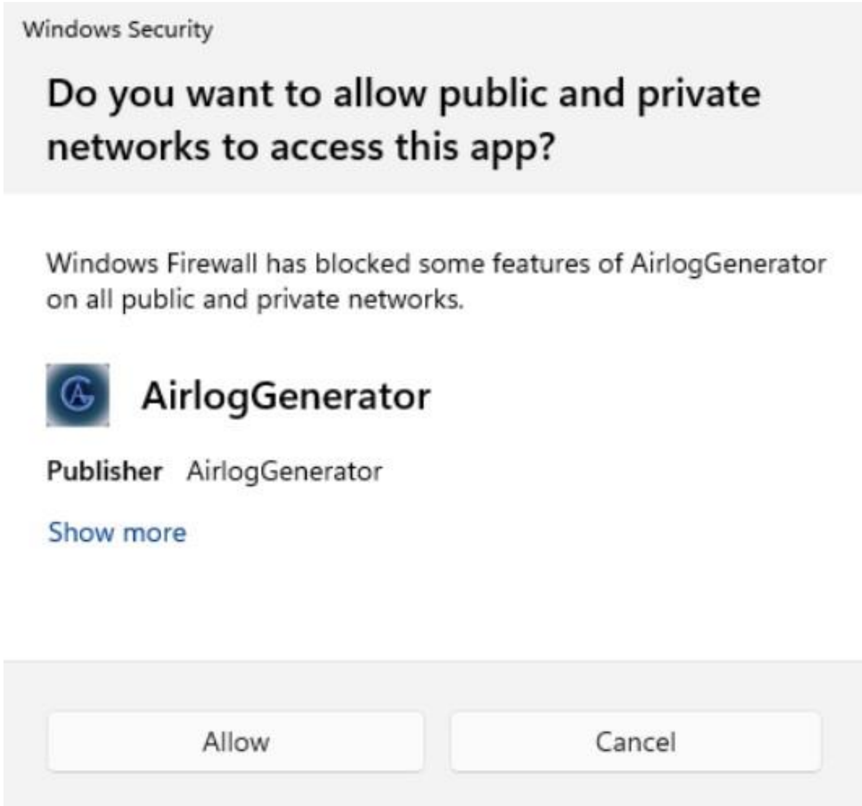


Running as a Standalone Application

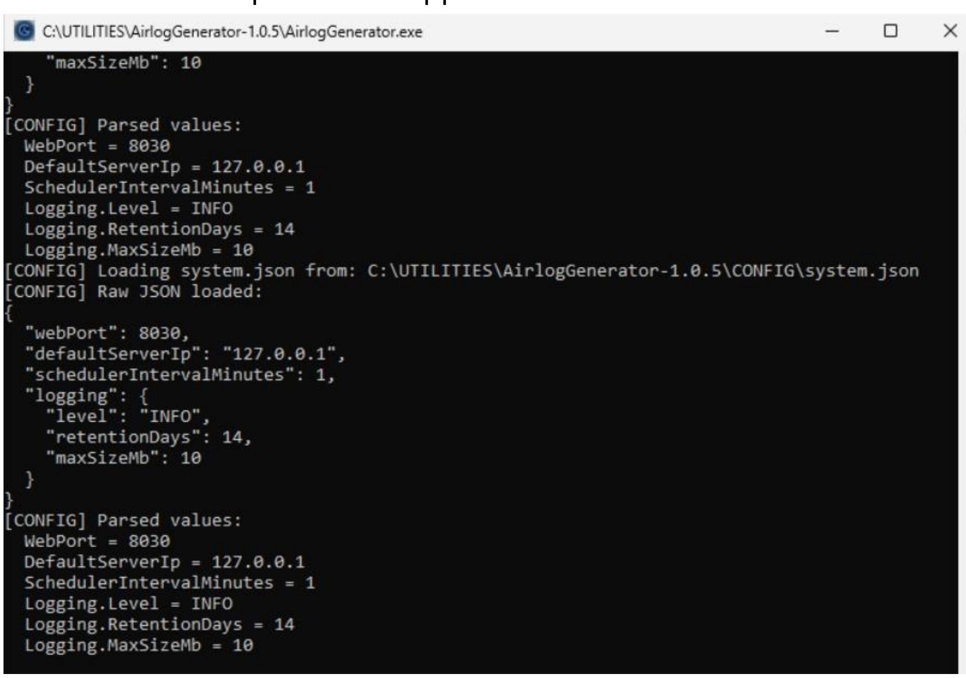
In the provided software package locate the AirlogGenerator.exe file. Right click on the file and “Run as Administrator”.

1) When the application is started, a dialogue box regarding access to network connections may be displayed. This allows the application to access the local network.

AIRLOG Generator



2) While the application is running, a terminal window will be displayed. The terminal window must remain open for the application to run.



Linux Installation

Getting Started

AIRLOG Generator has been compiled for use on a Linux Server. Copy the AG folder to the Home directory. The foler will contain ag-install.sh, docker-compoe, and the AirlogGenerator application folder. When installed it will create two symlinks:

1) /var/log/ag: This will house the application logs. By default a new log will start each day. The max log size for a day is 10 Mb, and will clear logs older than 14 days.
2) /etc/ag/: this will contain the system.json file that can be used for system settings for the application. A /stations/ directory will also be created. This will store the [station].json configuration file for each station profile.

Installing Docker Container

1) Change directories to ~/AG
2) Run the installer file to set the application directories
    - > docker compose build –no-cache
3) Run the docker up command
    - > docker compose up –d
4) Confirm the container is running
    - > docker compose ps

User Interface

The user interface can be accessed using a web browser. By default, the web service starts on port 8030. The port that is used can be modified as part of the system configuration.

System Settings

1) Open a web browser. Enter the IP address of the workstation where the application is running followed by a colon (:) and the configured port number (8030 by default). 127.0.0.1 or localhost can be used if accessing the interface from the same server where the application is running. (EX: http://127.0.0.1:8030)

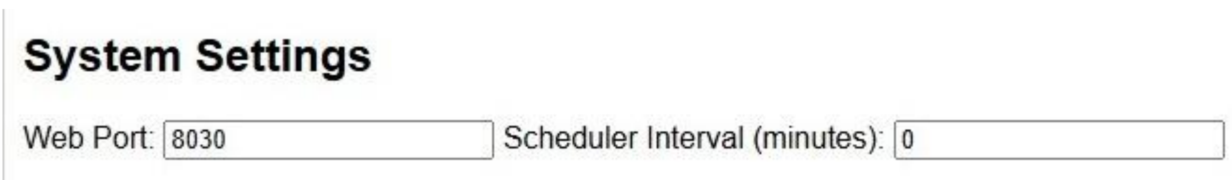
2) Across the top of the screen:
    a. The application version number will be displayed
    b. When connected to a server, the Central Server database version will be displayed.
    c. The WO AFR Software version will be displayed
    d. A Link to this manual will be available.

3) Enter in the IP address of the Central Server where the air logs will be generated from.
    a. This address will be saved to the station configuration files for the radio stations associated with the Central Server.
    b. It is possible to install AIRLOG Generator in a centralized location and connect to different Central Servers*

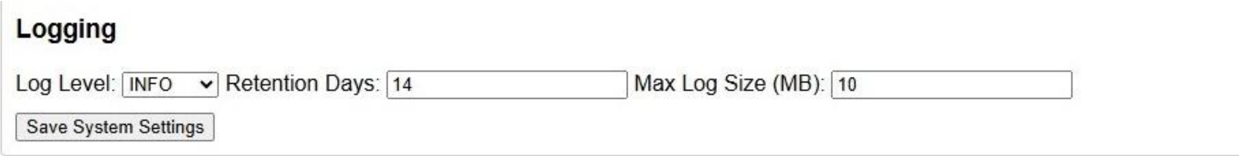
        * AIRLOG Generator connects to the Central Server database on port 5432. To use AIRLOG Generator from a remote location, it may be necessary to allow access to the port through the system firewall.

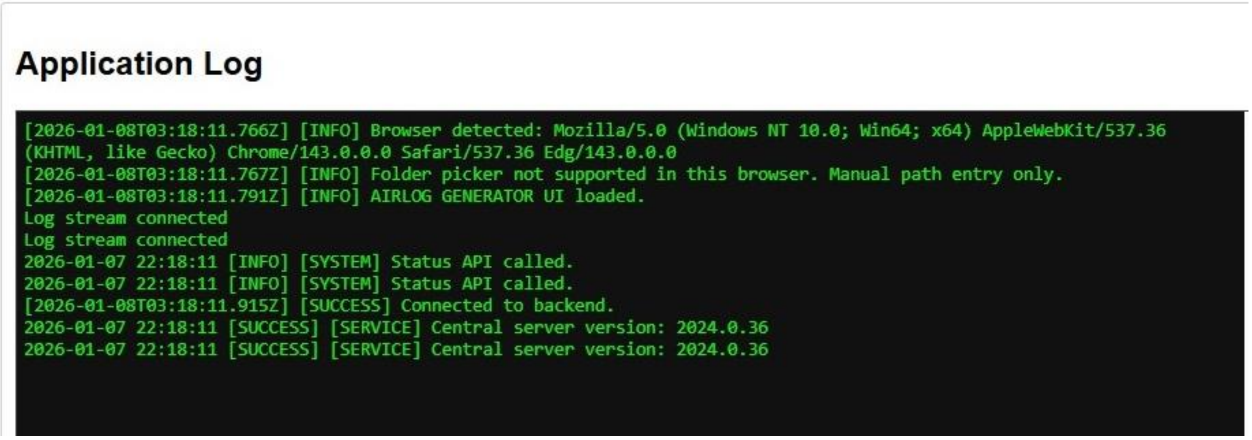
    c. Clicking connect will populate the container on the right side of the screen with a list of available radio stations from that server.

4) Web Port: defines the port used to access the User Interface. The default port is 8030. If this port is changed, it will require a restart of the service for the change to take effect.
5) Scheduler Interval: Define an interval will use to check for scheduled tasks.
   a. Setting an interval of 0 will keep the application into “Continuous Mode”. In this mode the service will check for tasks every second.
   b. Setting an interval value between 1 –120 minutes, the service will check for tasks at that interval.



6) AIRLOG Generator writes comprehensive application logs. A new log will be written to the LOG folder inside the application location. A new log will start each day.
   a. Log Level: Sets the level of logging that will be maintained. INFO is the default setting.
   b. Retention Days: Sets the number of days of logs will be retained. The default value is 14 days
   c. Max Log Size (MB): This sets the maximum log size. If a given day exceeds the maximum size, it will roll into a new log.
7) Application Log: Displays an active log window.
8) Save System Settings: Will write all setting changes to a system.json file located in a CONFIG folder within the application folder. Any changes made to the configuration will require a restart of the service after saving.
9) Station Configuration: When connected to a Central Server, this section will display a list of available radio stations associated with the connected Central Server.





Station Configuration

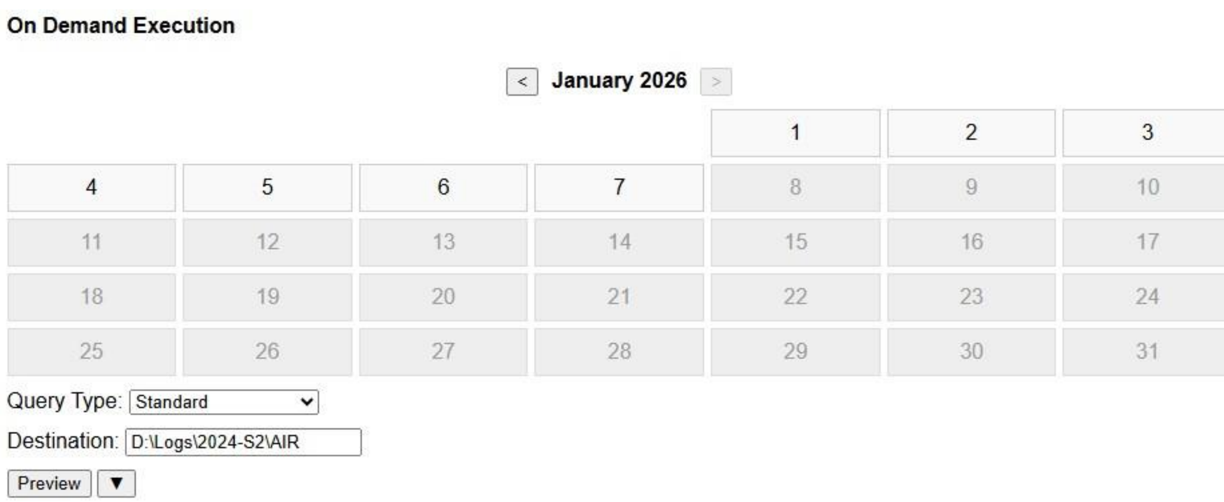
This section will display a list of radio stations associated with the connected Central Server. This initial display will show the radio station name as it is configured in the Central Server as well as a time zone offset that has been configured with the radio station. This section provides the ability to generate air logs on demand or configure a schedule to generate air logs.

1) Click on the desired Radio Station

2) Timezone Offset: Displays the offset configured on the radio station.
   a. The offset is NOT configurable from this application.
   b. Enable: when checked (default) the AIRLOG will be generated with adjusted times based on the offset*
      - In earlier version of the WO AFR software times were recorded based on the Central Server local time. This necessitated the need to adjust the time based on this offset. Starting with AFR2024, airtime was

recorded in the database based on the offset and further adjustment is not necessary. If running AFR2024 or later uncheck this box to disable the offset adjustments.

3) On Demand Execution: Allows the user to select one or more dates (only 30 days of logs are available). To select more than one date, hold down the Control Key and click on the desired dates.



a. Query Type provides options for the data provided in the air log
    - Standard – the standard air log. No rotator information is displayed
    - Partner ID – if a playlist (music or traffic) is processed using Partner ID, the artist field will be replaced with the Partner ID
    - Rotator Only – Provides an air log displaying only the Rotators Media Asset ID (MAID). The media asset that was played will NOT be listed.
    - Rotator + MAID – Will display the Rotator ID along with the MAID of the asset that played
    - Rotator + Partner ID – Displays the Rotator MAID and the Partner ID

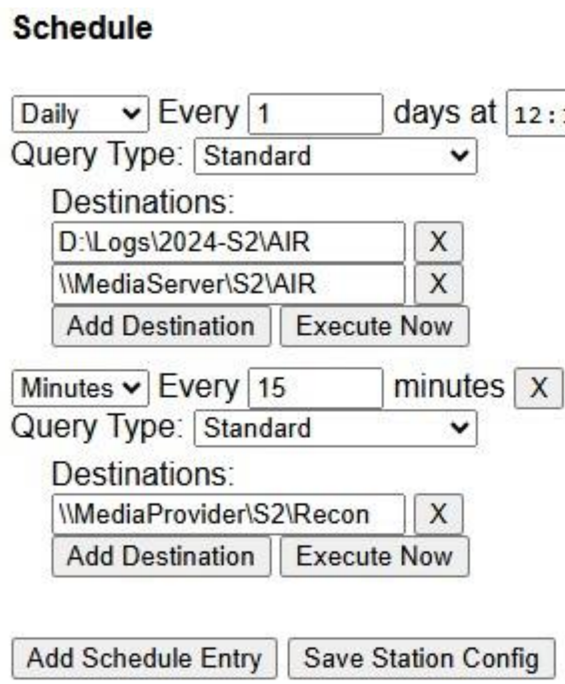
b. Destination: Set the destination where the AIRLOG will be written.

c. Multifunction Execution Button: Using the drop down a user will have the ability to set how the air logs will be presented

- Preview – Will display the selected air log(s) in a preview pane below within the UI
- Save / Write – will write the air log to the destination entered in the destination box
- Download – Downloads a copy of the file to the user's workstation.

Scheduled Tasks

While the service is running (both standalone and system service) allows a user to schedule tasks to be executed. Saving the Station Configuration will write these tasks along with the Host Central Server IP address, and Timezone offset election. While the service is running, the application will review the saved station.json files and execute the scheduled tasks against the Host Address, and Radio Station.



1) Add Scheduled Entry: add a scheduled task to be executed
2) Mode Dropdown box:
   a. Daily (Default) – Will be executed based on the configured number of days at the time specified. When executed it will generate logs from “yesterday” going back to the configured number of days.
   b. Minutes – Executes air log generation based on the configured number of minutes set. This interval can be affected if the Scheduler interval in the system setting is greater than the interval of the scheduled task.

c. Hourly - Executes air log generation based on the configured number of hours set. This interval can be affected if the Scheduler interval in the system setting is greater than the interval of the scheduled task.
d. Weekly – Scheduled tasks will be executed based on the number of weeks set on the day and time configured. It will generate a daily log for each day in the time frame set.
e. Monthly - Scheduled tasks will be executed based on the number of months set on the day and time configured. It will generate a daily log for each day in the time frame set. This setting should not be set higher than 1. Currently, the WO AFR system only retains 30 days of logs.

***Note: When setting the time to run the scheduled task the time is going to run based on the time zone of the workstation that the app is running on. For example, if the app is running on a Central Server in the Eastern Time zone and the task is set to run at 12:15am. The Radio Staion is operating in the Pacific Time zone, when the Central Server runs the task at 12:15am, it will only be 9:15PM for the radio station, the log will not be a complete day. ***

3) Add Destination: Allows the user to add destination(s) will be saved as part of the scheduled task.
4) Query type provides for setting the type of data that will be included in the scheduled task:
    i. Standard – the standard air log. Rotator information is not displayed
    ii. Partner ID – if a playlist (music or traffic) is processed using Partner ID, the artist field will be replaced with the Partner ID
    iii. Rotator Only – Provides an air log displaying only the Rotators Media Asset ID (MAID).
    iv. Rotator + MAID – Will display the Rotator ID along with the MAID of the asset that played
    v. Rotator + Partner ID – Displays the Rotator MAID and the Partner ID
5) Execute Now: Will execute the selected task immediately.
6) Save Station Configuration: Will write the schedule tasks to a station.json file. This file will also save host server address and timezone offset election.

AIRLOG Generator

System Configuration

It is possible for AIRLOG Generator to operate autonomously as a service or in a docker container. While running as a service, the application will rely on the system.json file and the [station].json files for the scheduled tasks.

AIRLOG Generator uses an unsecure web service for the UI to be active. If the server running the application is exposed directly to the internet, it may be undesirable for this service to be running. It is possible to stop the web service from starting in the system.json file.

For installtions that may be in an AWS environment where the Central Server's database has been seperated from the Central Server EC2 and loaded in an RDS environment, AIRLOG Generator can be configured to use a Secret Server passkey. This configuration is included in the System.json file. The use of this feature is optional. If not used, leave the section of the json file empty.

System.json File

The system.json file is stored in the /CONFIG folder of the application. This file can be edited with any text editor application.

This is the basic setup of the system.json file:

```json
{
  "webPort": 8030,
  "defaultServerIp": "127.0.0.1",
  "defaultServerHost": "",
  "defaultDatabaseName": "woar_server",
  "defaultServerUser": "",
  "defaultServerPassword": "",
  "defaultServerPasswordSecret": "",
  "defaultServerPasswordSecretRegion": "",
  "schedulerIntervalMinutes": 1,
  "logging": {
    "level": "INFO",
    "retentionDays": 14,
    "maxSizeMb": 10
  },
  "enableWebUi": true
}
```

<u>WebPort:</u> Default port for the webUI is 8030. This is the port used to access the webUI while the service is running. This is configurable via the webUI or by adjusting this file. A restart if the service is required for the change to take effect.

<u>DefaultServerIp:</u> This is the default IP used to connect to the Central Server. The default IP is adequate if the application is running on the same workstation as the Central Server Service. This can be changed and saved to the system.json file as needed. While using the webUI, the address set in this location is what will be written to the [station].json file. While using the webUI, the address set here will dictate the server used to gather the station information.

<u>defaultServerHost:</u> (Optional-Secret Server Configuration) Set this with the name or IP address of the RDS server where the database is stored.

<u>defaultDatabaseName</u>: (Optional-Secret Server Configuration) Leave this as "woar_server" unless your system has been specially configured with a different database name.

<u>defaultServerUser:</u> (Optional-Secret Server Configuration) Username to access the server.

<u>defaultServerPassword:</u> (Optional-Secret Server Configuration)

defaultServerPasswordSecret: (Optional-Secret Server Configuration)

<u>defaultServerPasswordSecretRegion:</u> (Optional-Secret Server Configuration)

<u>ScheduleIntervalMinutes:</u> While the service is running, this is the interval in which the application will check for new or pending tasks. If using Minutes or Hours for a scheduled task, note that this number should be lower than the interval set for the scheduled task. For example, if a scheduled task is set to run every 15 minutes, the Schedule Interval is set to 60 minutes. The scheduled task will run every 60 minutes when the service wakes up to check for new or pending tasks.

The value of 0 indicates that the service will never “sleep” and will check for tasks every second. The max value is 120 minutes.

<u>Level:</u> This is used to set the application level for logging. Options are Debug (verbose and detailed logging), Info (minimal operational logging), and Warn (only logging errors and warnings).

RetentionDays: This is the amount of time the application will store the application logs. This setting does not affect the air logs. By default, the system will retain system logs for 14 days.

MaxSizeMB: This is the maximum size a single system log can be. The default value is 10 Mb. The system will start a new log at the start of each day. If the daily log exceeds the value set here, it will roll to a new log YYMMDD-01.log.

EnableWebUI: this setting controls if the web service for the user interface starts. The web service used with this application is not secure. For workstations that may be exposed directly to the internet, it may be desirable to set this to False, preventing the web service from starting.

Station.json File

The application will store [station].json files in the /CONFIG/stations/ directory. This file contains the station with specific settings and schedules. While the service is running, the application will use this file to track the tasks to execute. The application will use the host IP address stored on this file to identify the Central Server to run the query for the scheduled task. This means it is possible for the single application to run tasks on more than one Central server.

The station configuration will be saved using the station name configured in Central Server. For example my station in Central Server is 2024-S2. The station json file is stored as 2024-S2.json.

Station.json Structure:

```
{
  "host": "192.168.50.150",
  "stationName": "2024-S2",
  "timezoneOffset": -3,
  "enableTimezoneCalc": false,
  "destinations": [],
  "schedule": [
    {
      "mode": "daily",
      "interval": 1,
      "time": "16:05",
      "day": null,
      "dayOfMonth": null,
      "destinations": [
        "\\\\192.168.50.150\\Logs\\Station2\\Airlog\\Daily"
```

```
      ],
      "queryType": 3
    },
    {
      "mode": "minutes",
      "interval": 15,
      "time": "00:15",
      "day": null,
      "dayOfMonth": null,
      "destinations": [
        "\\\\192.168.50.150\\Logs\\Station2\\Airlog\\min"
      ],
      "queryType": 2
    }
  ]
}
```

Host: The IP address of the Central Server the station resides on.

StationName: The radio station name as it is configured on the Central server

Timezoneoffset: The time zone offset between the radio station and the Central Server. When using the webUI this is a read only value. if set manually, make sure it matches the Radio Station Schedule configured in Central Server.

EnableTimezoneCalc: Setting this to "true" indicates that the application will automatically adjust the airtime recorded by the time zone offset. In older versions of WideOrbit Automation for Radio, the time recoded by the Central Server was based on the server's local time, and not the time zone the radio station was operating in. This necessitates the need to enable this setting for accurate time recording. Starting with AFR2024 the airtime recorded on the Central Server reflects the proper time zone of radio station operating. In the current versions of WO AFR this setting should be disabled.

Destinations: This is not used. no action is required.

Schedule: this section of the json files stores each of the scheduled tasks for this radio station. Each task will have a separate entry in the log.

- Mode: There are 5 available modes that can be set. Each sets a different frequency the schedule will be run
  - Daily: The task will run every configured number of days. This task will always start with "Yesterdays" log and go back from there. For example if the interval is set to 1, the task will run every day and generate Yesterdays log. If the interval is set to 3, the task will run every 3 days and will get yesterdays log and two back from there.
  - Hourly: will run based on the hour interval set. This task gets "Todays" log from this moment going back X hours.
  - Minute: will run based on the minute interval set. This task gets "Todays" log from this moment going back X minutes
  - Weekly: The task will run every X weeks generating logs from yesterday going back the set interval
  - Monthly: this setting should not be used.
- Interval: this sets the interval for the mode selected
- Time: if the mode requires it, this is the time of day the task will start.
- Day: If the mode requires it, this sets the day of the week the task will run
- DayOfMonth: Only use with the monthly mode.

Destinations: This allows for setting the specific destination or destinations the generated air logs should be saved.

QueryType: this allows for setting the specific query to run. there are five available options:

- 0) Standard: Produces the standard Airlog. It will not contain any rotator information, just the media asset that played.
- 1) PartnerID: If the music or traffic log was processed to use PartnerID, this will produce an air log where the artist field is replaced with the PartnerID
- 2) Rotator Only: Produces an air log where, for rotators, only the Rotator ID will be displayed. It will not contain the information for the media asset that is played.
- 3) Rotator + Media Asset: For rotators, this query will produce two log entries for rotators. Row 1 will be the scheduled rotator. Row 2 will be the media asset that ran.
- 4) Rotator + PartnerID: Produces a log where, if the music or traffic logs were processed using ParterID, where the rotator id and PartnerID will be displayed.